

Question 1 (14 marks) Start a New Booklet

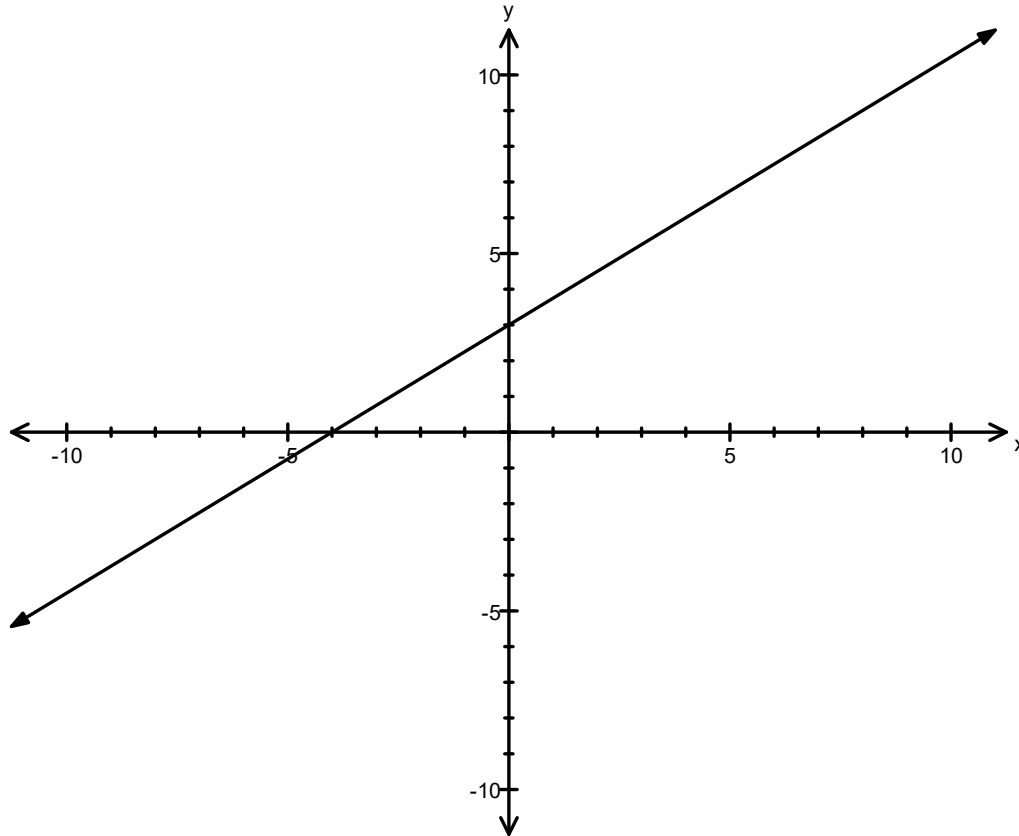
- (a) Find the value of $19^{-0.7}$ correct to two decimal places. 2
- (b) Find the integers a and b such that $(5 - \sqrt{2})^2 = a + b\sqrt{2}$. 2
- (c) Express $0.\dot{3}\dot{6}$ as a fraction in its simplest form. 2
- (d) Simplify $\frac{x}{3} + \frac{3x-1}{2}$ 2
- (e) Simplify $\frac{(3xy^3)^3}{3x^2y^4}$ 2
- (f) Factorise completely
- (i) $3x^3 + 24$ 2
- (ii) $x^4 - y^4$ 2

Question 2 (19 marks) Start a New Booklet

- (a) Solve simultaneously $\begin{cases} x + y = 1 \\ 2x - y = 5 \end{cases}$ 2
- (b) Solve $-4 \leq 2x - 3 \leq 7$ and graph your solution on the number line. 2
- (c) Given that $f(x) = \begin{cases} x+2 & \text{if } x < -1 \\ 1 & \text{if } -1 \leq x \leq 3 \\ x-2 & \text{if } x > 3 \end{cases}$ find $f(-2) + f(3)$. 2
- (d) Given the function $y = \sqrt{25 - x^2}$, state its domain and range. 2
- (e) Sketch, showing all essential features 4
- (i) $y = |x+2|$
- (ii) $y = \frac{1}{x+2} + 1$
- (f) Sketch the parabola $y = x^2 - x - 6$ indicating the x and y intercepts and vertex. 4
- (g) Solve, $3x^2 + 4x - 1 = 0$, leaving your answer in simplest form. 3

Question 3 (14 marks) Start a New Booklet

- (a) The line l cuts the x -axis at $M(-4, 0)$ and the y -axis at $N(0, 3)$ as shown. P is a point on the line l , and Q is the point $(0, 8)$.



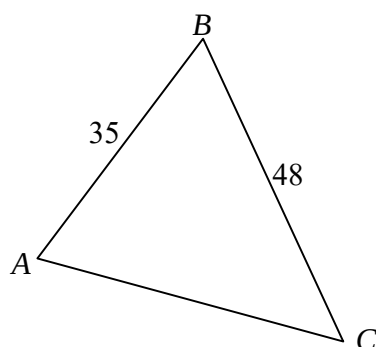
- | | | |
|--------|--|---|
| (i) | Copy the diagram into your writing paper, clearly marking the given coordinates. | |
| (ii) | Find the equation of the line l . | 2 |
| (iii) | Find the acute angle the line l makes with the x positive axis. | 1 |
| (iii) | Show that the point $(16, 15)$ lies on the line l . | 1 |
| (iv) | Show that $\triangle MNQ$ is isosceles. | 2 |
| (v) | Calculate the gradient of the line QM . | 1 |
| (vi) | N is the midpoint of the interval MP . Find the coordinates of the point P . | 2 |
| (vii) | Show that $\angle PQM$ is a right angle. | 1 |
| (viii) | Find the area of $\triangle PQM$. | 2 |

Question 3 cont.....

- (b) Find the perpendicular distance from the point (4, 5) to the line with the equation $5x - 12y - 4 = 0$. 2

Question 4 (16 marks) Start a New Booklet

- (a) Solve $2 \cos \theta + 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$. 2
- (b) Simplify $1 - \frac{\sin A \cos A}{\tan A}$. 3
- (c) In triangle ABC , $BC = 11\text{cm}$, $AC = 5\text{cm}$ and $AB = 8\text{cm}$.
- (i) Draw a diagram to show this information. 2
- (ii) Calculate the size of angle ABC , correct to the nearest degree. 2
- (iii) Calculate the area of the triangle (to nearest cm^2) 2
- (d) Three boys are standing on the school oval. A is 35 metres from B and B is 48 metres from C .
The bearing of B from A is 036°T and the bearing of C from B is 156°T .

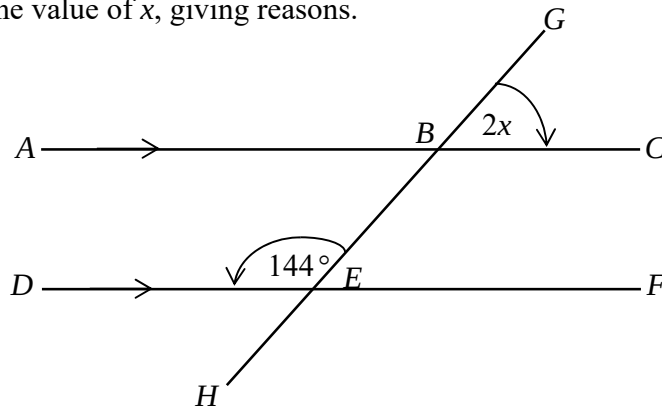


- (i) Copy the diagram into your answer booklet showing all information given above. 1
- (ii) Show that $\angle ABC = 60^\circ$. 1
- (iii) Find the distance of C from A , correct to the nearest metre. 2
- (iv) Find the bearing of A from C , correct to the nearest degree. 3

Question 5 (13 marks) Start a New Booklet

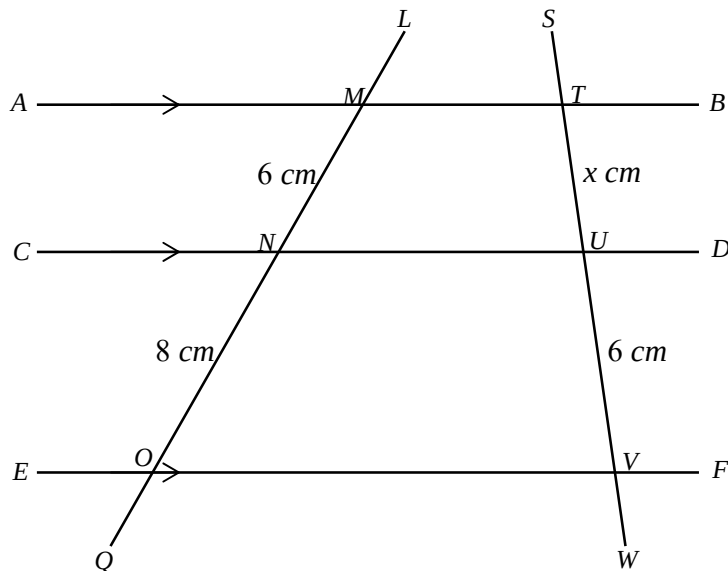
- a) Find the value of x , giving reasons.

2

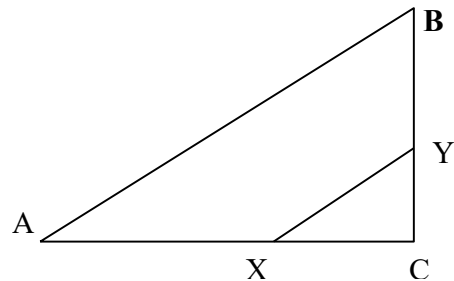


- (b) Find the value of x , give reasons to justify your answer.

2



- (c) In the figure XY is parallel to AB .
 $AB = 60$ mm
 $XY = 20$ mm
 $BY = 30$ mm



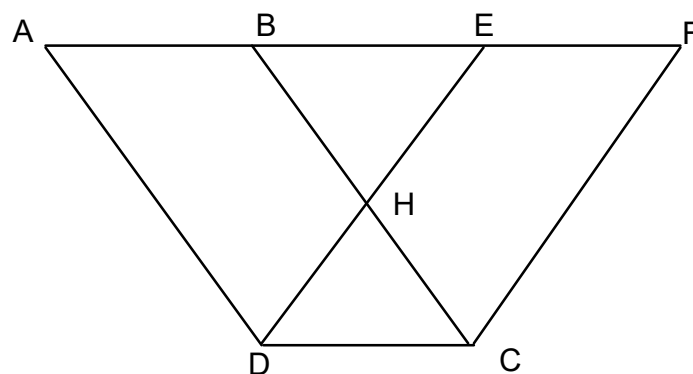
- (i) Show that $\triangle ABC$ is similar to $\triangle XYC$.
 (ii) Calculate the length of CY .

2

2

Question 5 cont...

(d)



Not To Scale

A, B, E and F are collinear points. ABCD and EFCD are parallelograms. BC and ED intersect at H such that H is the mid-point of BC. Copy or trace the diagram onto your worksheet.

- (i) Prove that $\triangle BHE \equiv \triangle CHD$. 3
- (ii) Explain why $DC = BE$. 1
- (iii) Hence or otherwise, show that $AF = 3DC$. 1

Question 6: (14 marks) Start a New Booklet

Differentiate the following, with respect to x

- (i) $4x^3 - 6x + 7$ 1
- (ii) $\frac{2}{x^3}$ 2
- (iii) $x\sqrt{x}$ 2
- (iv) $5x(2x-1)^2$ 2
- (v) $(4x^3 - 7)^5$ 2
- (vi) $\frac{4x^2 - 2}{x + 3}$ 2
- (vii) $x\sqrt{x+3}$, answer in its simplest form. 3

END OF THE PAPER

$$a) \quad 19^{-0.7} = 0.127 \quad (1 \text{ mark}) \quad (6/10) \\ = \underline{\underline{0.13}} \quad (2)$$

$$b) \quad (5 - \sqrt{2})^2 = 25 - 10\sqrt{2} + 2 \\ = 27 - 10\sqrt{2} \quad (1 \text{ mark}) \quad (2) \\ \therefore a = 27, b = -10 \quad (2)$$

$$c) \quad 0.\dot{3}\dot{6} = \frac{36}{99} \quad (1 \text{ mark}) \\ = \frac{4}{11} \quad (2)$$

$$d) \quad \frac{x}{3} + \frac{3x-1}{2} \\ = \frac{2x + 3(3x-1)}{6} \quad (1 \text{ mark}) \\ = \frac{2x + 9x - 3}{6} \\ = \frac{11x - 3}{6} \quad (2)$$

$$e) \quad \frac{(3xy^3)^3}{3x^2y^4} = \frac{27x^3y^9}{3x^2y^4}$$

$$= \underline{\underline{9xy^5}} \quad (2)$$

$$f) \quad 3x^3 + 24 \\ = 3(x^3 + 8) \quad (1 \text{ mark}) \\ = \underline{\underline{3(x+2)(x^2 - 2x + 4)}} \quad (2)$$

$$ii) \quad x^4 - y^4 \\ = (x^2 - y^2)(x^2 + y^2) \quad (1 \text{ mark}) \\ = \underline{\underline{(x+y)(x-y)(x^2 + y^2)}} \quad (2)$$

Preliminary Examination.

2004 Year 11 2010

Question 2

$$1. \quad x + y = 1 \quad \text{--- (1)}$$

$$2x - y = 5 \quad \text{--- (2)}$$

$$1 + (2) \quad 3x = 6$$

$$\therefore x = 2 \quad \checkmark$$

$$\text{Sub in (1)} \quad 2 + y = 1$$

$$y = -1$$

2 marks

$$\text{check in (2)} \quad 4 - (-1) = 5 \checkmark$$

$$\therefore x = 2, y = -1 \text{ is the solution.} \quad \checkmark$$

(2 marks).

$$b) \quad -4 \leq 2x - 3 \leq 7$$

$$-1 \leq 2x \leq 10$$

$$-\frac{1}{2} \leq x \leq 5 \quad \checkmark$$

2 marks

Solutions should be connected to number line



(2 marks)

$$c) \quad \left. \begin{array}{l} f(-2) = 0 \\ f(3) = 1 \end{array} \right\} \begin{array}{l} \text{1 mark for} \\ \text{one of} \\ \text{these} \end{array} \quad \checkmark$$

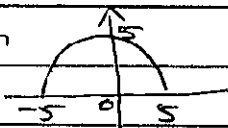
2 marks

$$\therefore f(-2) + f(3) = 0 + 1 = 1 \quad \checkmark$$

(2 marks)

sketch

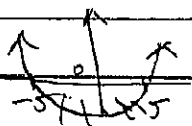
$$d) \quad y = \sqrt{25 - x^2}$$



$$\text{Domain: } 25 - x^2 \geq 0$$

$$(5 - x)(5 + x) \geq 0$$

2 marks



$$\text{Domain: } -5 \leq x \leq 5 \quad \checkmark$$

$$\text{Range: } 0 \leq y \leq 5 \quad \checkmark$$

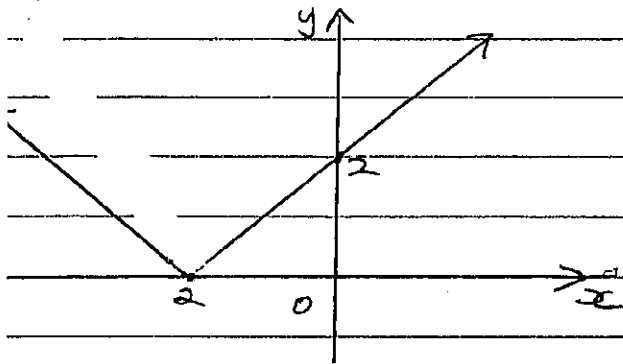
(2 marks)

Domain + Range must be named accordingly.

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2 unit Year 11 2010.

(e)(i) $y = |x+2|$

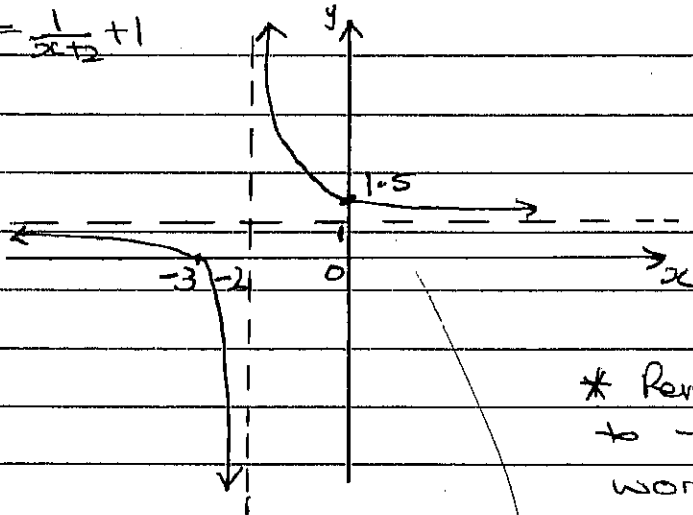


1 mark both intercepts.

1 mark shape.

2 marks

(ii) $y = \frac{1}{x+2} + 1$



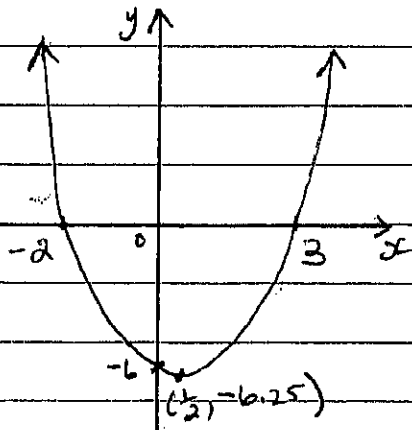
1 mark shape

1 mark all other details correct.

2 marks

* Remember to add the information to the graphs. Some students worked out information near the graphs but forgot to put in on the solution.

(f) $y = x^2 - x - 6$



Roots $x^2 - x - 6 = 0$

$(x-3)(x+2) = 0$ 1 mark

$\therefore x = -2, 3$

y-int $= -6$

1 mark

Vertex $= (\frac{1}{2}, -6.25)$

1 mark

Shape of graph

1 mark

4 marks

(g) $x = \frac{-4 \pm \sqrt{4^2 - (4 \times 3 \times -1)}}{2 \times 3}$ ✓ 1 mark

$= \frac{-4 \pm \sqrt{28}}{6}$

3 marks

$= \frac{-4 \pm 2\sqrt{7}}{6}$ ✓ 1 mark

$x = \frac{-2 \pm \sqrt{7}}{3}$ ✓ 1 mark

Q3 - 2 unit

ii) $m = \frac{3}{4}$ $b = 3$ (1 mark)

$\therefore y = \frac{3}{4}x + 3$ (2)

iii) $\tan^{-1}\left(\frac{3}{4}\right) = 36.869... (calc)$

$\therefore \text{Angle } C \hat{=} 37^\circ$ (1)

iii) Sub (16, 15) into $y = \frac{3}{4}x + 3$

$15 = \frac{3}{4} \times 16 + 3$

$15 = 15$ $\therefore$ Point lies on line. (1)

iv) $PN = 5$ (1)

$MN^2 = 3^2 + 4^2$
 $= 25$

$MN = \sqrt{25}$

$MN = 5$ (1)

$\therefore$ Since 2 Sides =

ΔMNP is Isosceles

v) $M_{QM} = \frac{8}{4}$ (1)
 $= 2$

vi) $M(-4, 0)$ $N(0, 3)$

$\frac{x+4}{2} = 0$

$\frac{y+0}{2} = 3$

$\therefore P = (4, 6)$ (2)

vii) $m_{PQ} = -\frac{2}{4}$ $m_{QM} = 2 (m_1)$

$m_1 = -\frac{1}{2}$

Since $m_1 \times m_2 = -1$ (1)

then $PQ \perp QM$

$\therefore \angle PQM = 90^\circ$

viii) $QM = \sqrt{80}$

$PQ = \sqrt{20}$

$A = \frac{1}{2} \times 4 \times 8 + \frac{1}{2} \times 2 \times 4$
 $= 20 \text{ units}^2$

$\therefore A = \frac{1}{2} \times \sqrt{80} \times \sqrt{20}$
 $= 20 \text{ units}^2$

(2)

b) $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

$= \frac{|4 \times 5 + 5 \times -12 - 4|}{\sqrt{5^2 + (-12)^2}}$

$= \frac{44}{\sqrt{169}}$

$= \frac{44}{13} \text{ units}$

(2)

a) $2 \cos \theta = -1$
 $\cos \theta = -\frac{1}{2}$
 $(\cos 60 = \frac{1}{2})$ (2)
 $\therefore \theta = 120^\circ, 240^\circ$ for both

if $\theta = 120^\circ$ is given with

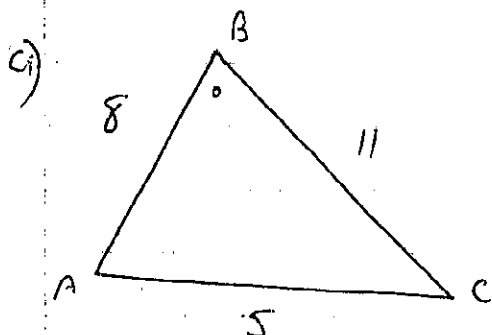
b) $1 - \frac{\sin A \cos A}{\tan A}$ no related angle

$$= 1 - \sin A \cos A \div \frac{\sin A}{\cos A}$$

$$= 1 - \sin A \cos A \times \frac{\cos A}{\sin A}$$

$$= 1 - \cos^2 A$$

$$= \sin^2 A \quad (3)$$



$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$= \frac{11^2 + 8^2 - 5^2}{2 \times 8 \times 11}$$

$$= 0.909 \dots \text{ (Calc)}$$

$$\angle ABC = 24.6 \dots$$

$$= 25^\circ \quad (2)$$

ii) $A = \frac{1}{2} cb \sin C$

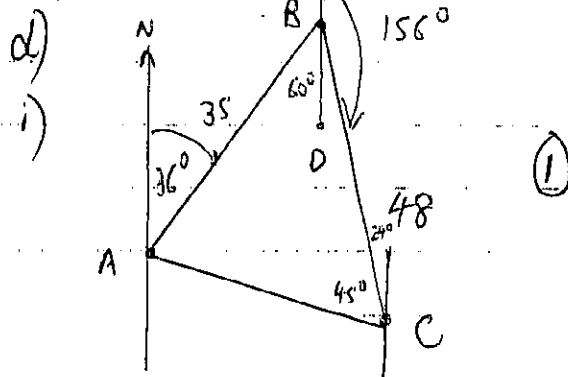
$$= \frac{1}{2} \times 8 \times 11 \times \sin 25$$

$$= 18.59 \dots$$

$$= 19 \text{ cm}^2$$

Any angle
can get
2 marks

(2) if the complete calculator answer not accept rounded answer was used
Bearing = 291 (3)



i) $\angle ABD = 36^\circ$ (Alt L's on $\parallel$)

$$\angle CBD + 156 = 180 \text{ (Straight L)}$$

$$\angle CBD = 24^\circ$$

$$\angle ABC = 24 + 36 = 60^\circ$$

(Find 1 mark)

iii) $a^2 = b^2 + c^2 - 2bc \cos A$
 $= 35^2 + 48^2 - 2 \times 35 \times 48 \cos 60$

$$a^2 = 1849$$

$$a = \sqrt{1849}$$

$$a = 43$$

$$\therefore AC = 43 \text{ m}$$

iv) $\frac{c}{\sin C} = \frac{b}{\sin B}$

$$\frac{\sin C}{35} = \frac{\sin 60}{43}$$

$$\sin C = \frac{35 \sin 60}{43}$$

$$\sin C = 0.7046 \dots$$

$$\angle C = 44.8 \dots$$

$$\angle C = 45^\circ$$

$$\text{Bearing} = 291 \quad (2 \text{ marks})$$

Question 5

3 marks

- (i) In $\triangle BHE$ and $\triangle CHD$
 $BH = CH$ (H is the midpoint of BC) ✓ (NOT GIVEN) *
 $\angle BHE = \angle CHD$ (vertically opposite angles) ✓
 $\angle EBH = \angle DCH$ (alternate angles on parallel lines) ✓
 $\therefore \triangle BHE \equiv \triangle CHD$ (A.A.S.) ✓

1 mark 1 part correct for reasons

2 marks 2 parts correct for reasons

3 marks Fully correct.

- (ii) $DC = BE$ (corresponding sides in congruent triangles) ✓ (1 mark)

- (iii) $DC = BE$ (from (ii))

$AB = DC$ (opposite sides in parallelogram)

$EF = DC$ (opposite sides in parallelogram)

$$AB + BE + EF = 3 \times DC$$

$$\therefore AF = 3DC$$

1 mark

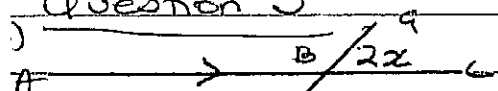
Fully Correct solution only

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2 unit Year 11 2010

Question 5

(2 marks)



$\angle ABE = 2x^\circ$ (vertically opposite angle)

$2x^\circ + 144^\circ = 180^\circ$ (co-int. angles on parallel lines)

OR
(co-int. angles, $AC \parallel DF$)

$$2x^\circ = 18^\circ$$

* Poor use of reasons. Inappropriate use of symbols.

$\angle$ for angle may be used.

* If reasons incomplete or non-existent 1 mark given for 180

(b) $\frac{6}{8} = \frac{2x}{6}$ (intercepts on parallel lines) OR intercept theorem

$$8x = 36$$

$$x = 4.5 \text{ cm}$$

2 marks

(c)(i) In $\triangle ABC$ and $\triangle XYC$

$\angle C$ is common

$\angle ABC = \angle XYC$ (corresponding angles, $AB \parallel XY$)

$\angle BAC = \angle YXC$ (corresponding angles, $AB \parallel XY$)

} only 2 angles are necessary

$\therefore \triangle ABC \sim \triangle XYC$ (equiangular) OR
(all pairs of corresponding angles are equal)

* Proofs must be finished off properly. Lazily done

(ii) $\frac{x}{x+30} = \frac{20}{60}$ (corresponding sides in similar triangles are in proportion)

$$60x = 20(x+30)$$

$$60x = 20x + 600$$

$$40x = 600$$

$$x = 15$$

$$\therefore CY = 15 \text{ mm}$$

* A reason must be given.

i) $y = 4x^3 - 6x + 7$

$y' = 12x^2 - 6$ (1)

ii) $y = \frac{2}{x^3}$

$y = 2x^{-3}$

$y' = -6x^{-4}$ (Accept for 2 marks)

$y' = \frac{-6}{x^4}$ (2)

iii) $y = x\sqrt{x}$

$y = x \times x^{\frac{1}{2}}$

$y = x^{\frac{3}{2}}$

$y' = \frac{3}{2}x^{\frac{1}{2}}$ (Accept) (2)

$y' = \frac{3\sqrt{x}}{2}$

iv) $y = 5x(2x-1)^2$

$y = 5x(4x^2 - 4x + 1)$

$y = 20x^3 - 20x^2 + 5x$

(2) $y' = 60x^2 - 40x + 5$

$u = 5x \quad v = (2x-1)^2$
 $u' = 5 \quad v' = 2(2x-1) = 4(2x-1)$

$y' = vu' + uv'$
 $= (2x-1)^2 \cdot 5 + 5x \cdot 4(2x-1)$
 $= (4x^2 - 4x + 1)5 + 20x(2x-1)$
 $= 20x^2 - 20x + 5 + 40x^2 - 20x$
 $= 60x^2 - 40x + 5$

$5(2x-1)(2x-1+4x)$
 $= 5(2x-1)(6x-1)$

v) $y = (4x^3 - 7)^5$

$y' = 5 \times 12x^2(4x^3 - 7)^4$ (2)

$y' = 60x^2(4x^3 - 7)^4$

vi) $u = 4x^2 - 2 \quad u' = 8x$

$v = x+3 \quad v' = 1$

$y' = \frac{vu' - uv'}{v^2}$

$y' = \frac{(x+3) \cdot 8x - (4x^2 - 2) \cdot 1}{(x+3)^2}$

$y' = \frac{8x^2 + 24x - 4x^2 + 2}{(x+3)^2}$

$y' = \frac{4x^2 + 24x + 2}{(x+3)^2}$ (2)

$y' = \frac{2(2x^2 + 12x + 1)}{(x+3)^2}$

vii) $u = x \quad v = (x+3)^{\frac{1}{2}}$
 $u' = 1 \quad v' = \frac{1}{2}(x+3)^{-\frac{1}{2}}$

$y' = vu' + uv'$
 $y' = (x+3)^{\frac{1}{2}} + x \cdot \frac{1}{2}(x+3)^{-\frac{1}{2}}$
 $= (x+3)^{\frac{1}{2}} + \frac{x}{2(x+3)^{\frac{1}{2}}}$

$= \frac{2(x+3) + x}{2(x+3)^{\frac{1}{2}}}$

$= \frac{2x + 6 + x}{2\sqrt{x+3}} = \frac{3x+6}{2\sqrt{x+3}} = \frac{3(x+2)}{2\sqrt{x+3}}$ (3)